Finite Nonlinear Separations for Exact Four-Cycle Graph Codes

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Abstract

Identify labelled graphs on $[n]$ with vectors of $\mathbb{F}_2^{\binom{n}{2}}$, and prohibit pairs whose symmetric difference is *exactly* the edge set of a four-cycle, with all remaining edges absent. We determine the maximum linear code size for every $n \geq 4$, namely $2^{\binom{n}{2}-\lceil \log_2 n \rceil}$. At $n = 5$ the unrestricted maximum is 192, whereas the linear maximum is 128. We give an explicit small clique-cover certificate for the unrestricted bound and a construction attaining it. We further prove the exact six-vertex value $D_{C_4}(6) = 6144 > 4096 = D^{\mathrm{lin}}_{C_4}(6)$ by a finite Fourier-positive kernel certificate and a reproducible matching construction. This note separates mathematical verification from the still unresolved question of publication priority. A 2026 survey already gives nonlinear-versus-linear gaps for broader forbidden *families*; no claim of the first such gap in general is made here.

1 Definitions and context

Put $m = \binom{n}{2}$. Represent a graph on $[n]$ by its edge-indicator vector $x \in \mathbb{F}_2^m$. Let $F_n \subseteq \mathbb{F}_2^m$ be the set of $3\binom{n}{4}$ vectors whose nonzero coordinates form exactly the four edges of a labelled C_4. A set $A \subseteq \mathbb{F}_2^m$ is C_4-admissible when $x + y \notin F_n$ for every distinct $x, y \in A$. Write

$$D_{C_4}(n) = \max_{A\ C_4\text{-admissible}} |A|, \qquad D^{\mathrm{lin}}_{C_4}(n) = \max_{\substack{A \leq \mathbb{F}_2^m \\ A \cap F_n = \varnothing}} |A|.$$

This is the exact-copy problem introduced in Alon's graph-code framework [1]; it is *not* the stricter problem that forbids differences merely *containing* a four-cycle. Linear variants and associated odd edge-colouring ideas have been studied in [2, 3].

2 The linear optimum

Theorem 1. *For every $n \geq 4$,*

$$\boxed{D^{\mathrm{lin}}_{C_4}(n) = 2^{\binom{n}{2}-\lceil \log_2 n \rceil}.}$$

For $n \leq 3$, no C_4 occurs and the unrestricted graph space is admissible.

Proof. A parity-check map $\mathbb{F}_2^m \to \mathbb{F}_2^r$ is described by edge labels $c_{uv} \in \mathbb{F}_2^r$, and its kernel avoids F_n precisely when the sum of the labels around each C_4 is nonzero. The construction is immediate from a field: let $q = 2^{\lceil \log_2 n \rceil}$, choose distinct $x_1, \ldots, x_n \in \mathbb{F}_q$, and put $c_{ij} = x_i x_j$. For a cycle (a, b, c, d, a) the sum equals

$$(x_a + x_c)(x_b + x_d) \neq 0.$$

Thus a parity-check map with at most $\lceil \log_2 n \rceil$ bits exists.

For the lower bound, first $r = 0$ is impossible on K_4. If $r = 1$, the three distinct four-cycles on a fixed K_4 have edge sets whose XOR is zero. All three would have to map to the unique nonzero value 1, an impossibility. Now let $r \geq 2$ and put $q = 2^r$. Suppose K_{q+1} has edge labels $c_{uv} \in \mathbb{F}_2^r$ with nonzero sum on every four-cycle. For each vertex define $s_u = \sum_{v \neq u} c_{uv}$ and switch labels by

$$c'_{uv} = c_{uv} + s_u + s_v.$$

Four-cycle sums are unchanged. Since q is even and $\sum_u s_u = 0$, the sum of the q switched labels incident with any vertex is zero.

Fix distinct vertices u, v. For their $q - 1$ other neighbours w, set $d_w = c'_{uw} + c'_{vw}$. Any two d_w are distinct: equality would yield a four-cycle of label sum zero. Also $\sum_{w \neq u,v} d_w = 0$, by the zero incident sums at u, v. The XOR of *all* q elements of $\mathbb{F}_2^r$ is zero when $r \geq 2$. Consequently, the one vector missing from the $q - 1$ distinct values d_w must be zero. Therefore $d_w \neq 0$ for every w.

It follows that any two edges incident with a common vertex have different labels. Every vertex of K_{q+1} has degree q, and there are precisely q labels: every label occurs on exactly one edge incident with every vertex. The edges of any fixed label would form a perfect matching of K_{q+1}, contradicting its odd order. Thus K_{q+1} is impossible with r bits. Restricting any labelling on K_n to $q + 1$ vertices shows that $n \leq 2^r$. Hence $r \geq \lceil \log_2 n \rceil$. □

3 An exact nonlinear separation on five vertices

Theorem 2. *For the exact-C_4 forbidden-difference problem,*

$$\boxed{D_{C_4}(5) = 192 > 128 = D^{\text{lin}}_{C_4}(5).}$$

Proof. Order the ten coordinates as $01, 02, 03, 04, 12, 13, 14, 23, 24, 34$ (with vertices $0, \ldots, 4$), and interpret bit masks as integers. The fifteen forbidden C_4 masks span a five-dimensional subspace $S \leq \mathbb{F}_2^{10}$; for example, the forbidden masks $54, 90, 149, 281, 553$ are linearly independent. The span is exactly the space of even-degree, even-edge-count graphs: its five-dimensionality follows from the four independent vertex-degree parity constraints and one additional independent edge-parity constraint. In particular $\mathbb{F}_2^{10}$ splits into $2^5 = 32$ cosets of S, and forbidden differences connect elements within the same coset only.

Any admissible set inside a coset may be translated to contain 0. Its further members must avoid the fifteen forbidden masks. There are precisely sixteen eligible nonzero masks, and they admit the following partition into five *conflict cliques* (each pair within a row has forbidden XOR):

207	303	543	1023
249	373	627	
442	470	915	
700	742	933	
860	874	969	

Thus at most one additional member comes from each row, for at most $1 + 5 = 6$ codewords per coset. The six masks

$$B = \{0, 207, 373, 700, 874, 915\}$$

indeed have no forbidden pairwise difference. Taking a translated copy of B in each of the 32 cosets gives $32 \cdot 6 = 192$ graph-codewords; differences across different cosets cannot be forbidden. Therefore the upper bound is attained. The linear figure 128 follows from Theorem 1. □

Remark 1. *The conflict graph on the fifteen eligible masks other than* 1023 *is isomorphic to the Kneser graph* $KG(6,2)$*, while* 1023 *conflicts with all fifteen. This provides a conceptual interpretation of the five-row partition; the explicit partition and witness suffice for the proof.*

4 An exact six-vertex separation

Theorem 3. *For the exact-*C_4 *forbidden-difference problem,*

$$\boxed{D_{C_4}(6) = 6144 > 4096 = D^{\mathrm{lin}}_{C_4}(6).}$$

Proof. Let F denote the 45 forbidden four-cycle vectors in $V = \mathbb{F}_2^{15}$, and let $S = \langle F \rangle$. Elementary binary elimination gives $\dim S = 9$; hence V is a disjoint union of 64 conflict-isomorphic cosets of S. It suffices to prove that the Cayley conflict graph $\mathrm{Cay}(S, F)$ has independence number 96. A 96-word admissible seed in S is supplied by the machine-readable witness `no26_n6_improved_witness_6144.json`. Its validity is checked by the independent code in the companion evidence.

For the matching upper bound, let $T \subset S$ be the ten edge sets that are disjoint unions of two triangles partitioning the six vertices. Define

$$K(x) = 96\mathbf{1}_{\{0\}}(x) - 1 + \frac{32}{3}\mathbf{1}_F(x) - \frac{32}{5}\mathbf{1}_T(x), \qquad x \in S.$$

Then $K(0) = 95$, and $K(x) \le -1$ for every $x \neq 0$ outside F. We claim K is positive definite. For a character $\chi : S \to \{\pm 1\}$, write $a_\chi = \sum_{f \in F} \chi(f)$ and $b_\chi = \sum_{t \in T} \chi(t)$. For the trivial character, $\widehat{K}(1) = 96 - 512 + (32/3)45 - (32/5)10 = 0$. For each nontrivial character,

$$\widehat{K}(\chi) = \frac{32}{15}(45 + 5a_\chi - 3b_\chi).$$

The full joint spectrum is enumerated in the table below. Every entry in the last column is nonnegative, proving $\widehat{K}(\chi) \ge 0$ for all characters. The last row is the trivial character, whose actual Fourier value is 0 as computed separately. These integers can be reproduced by enumerating the 2^{15} ambient edge-parity characters, each of whose restrictions to S occurs 2^6 times.

a_χ	b_χ	$\#\chi$	$45 + 5a_\chi - 3b_\chi$	a_χ	b_χ	$\#\chi$	$45 + 5a_\chi - 3b_\chi$
−15	−10	6	0	−7	−2	90	16
−3	2	180	24	−3	10	15	0
1	−2	90	56	5	2	45	64
9	−10	10	120	9	−2	60	96
21	2	15	144	45	10	1	240

For any F-admissible $A \subseteq S$ of size M, Fourier positivity yields

$$0 \le \sum_{u,v \in A} K(u+v) = \frac{1}{512}\sum_{\chi} \widehat{K}(\chi)\left|\sum_{u \in A}\chi(u)\right|^2 \le 95M - M(M-1) = M(96-M).$$

Therefore $M \le 96$. The verified seed attains 96, and independent translations to the 64 cosets of S give $64 \cdot 96 = 6144$ words. The linear figure follows from Theorem 1. □

5 Scope and priority status

This draft reports exact finite results at $n = 5, 6$ and a linear all-n formula. No asymptotic separation for a fixed C_4 is claimed. In particular, Alon's 2026 survey [4] presents already-known substantial nonlinear-versus-linear gaps for *families* of forbidden graphs, so nonlinear

superiority per se is not a new observation. The survey was located in full text during this audit; the specific singleton-C_4, five-vertex numerical statement was not found in the reviewed passages. Absence in those passages is not a proof of publication priority. The all-n parity-check formula likewise requires historical comparison with zero-sum edge-labelling literature. **This manuscript is a review draft, not a claim of first discovery or a refereed publication.**

Prior-art inquiries. To the best of the author's knowledge, the exact values $D_{C_4}(5) = 192$, $D_{C_4}(6) = 6144$ and the linear all-n formula in this note were not found in the literature examined. This statement is limited to the sources reviewed and does not establish priority. Readers aware of earlier or overlapping results are invited to contact Yonezu Research Lab at yonezaemon1+research@gmail.com.

Reproducibility. A standard-library Python script independently reconstructs all forbidden masks, checks the five conflict cliques and the six-word seed, and verifies the 192-word expanded code. Two further integer-only verifiers check the six-vertex positive-definite spectral certificate in different coordinate systems, and reconstruct the 6144-word witness with all pairwise differences. The accompanying reproducibility record contains SHA-256 hashes for the input witness, verification certificates, and proof scripts. The exact six-vertex Fourier spectrum, kernel conditions, and matching construction have also been independently recomputed using a separate implementation. This does not constitute professional peer review; historical priority remains unresolved.

References

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[4] N. Alon, *Graph-codes: questions, results and methods*, Japanese J. Math. **21** (2026), 145–164. https://doi.org/10.1007/s11537-026-2536-1. Author-hosted full text: https://web.math.princeton.edu/~nalon/PDFS/gcs4.pdf.